

Jellyfin Media Server

Home Lab Project Documentation

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Project Overview

This project documents the installation, configuration, and evaluation of Jellyfin, a free, open-source media server deployed on my home Windows server as a direct alternative to Plex Media Server. The core motivation was to assess whether an open-source solution could match or exceed Plex's feature set without the cost barrier of a Plex Pass subscription.

Key outcomes of this project include:

- Fully functional Jellyfin server with anime library integration
- Hardware-accelerated transcoding via Intel ARC B570 (Intel QSV / Quick Sync)
- Custom UI theming using community CSS from GitHub
- Plugin-based enhancements: Media Bar Enhanced & Intro Skipper
- Secure remote access via Tailscale VPN tested on an iPhone 16 Pro over cellular

Motivation & Goals

Plex Media Server is a well-established media platform but gathers a significant number of its most useful features behind a paid subscription called Plex Pass, which costs between \$6.99/month and a \$249 lifetime purchase. Features like hardware transcoding and a Skip Intro button which are fundamental to a smooth streaming experience are unavailable on the free tier.

Jellyfin, by contrast, is entirely open-source and community-maintained with no paywalled features. This project was initiated to:

- Evaluate Jellyfin as a cost-free drop-in alternative to Plex
- Test parity of features, particularly those locked behind Plex Pass
- Explore the open-source plugin and theming ecosystem
- Practice hardware transcoding configuration with Intel ARC / QSV
- Expand home lab documentation for portfolio purposes

System Environment

Component	Specification
OS	Windows (Home Server)
CPU	AMD Ryzen 5 5500
GPU	Intel ARC B570 (Hardware Transcoding via Intel QSV)
Media Server	Jellyfin (installed via .exe)
VPN / Remote Access	Tailscale (Open-Source)
Test Device	iPhone 16 Pro (cellular remote access test)

Build Process

Step 1 — Download & Install Jellyfin

Downloaded the Jellyfin Windows installer (.exe) from the official Jellyfin website and ran a standard installation on the home server. No command-line setup was required at this stage.

Step 2 — Initial Setup: Admin Account & Remote Access

Completed the Jellyfin first-run wizard, which included:

- Creating an admin account with a secure username and password
- Enabling remote access within Jellyfin's network settings to allow connections beyond the local network

Step 3 — Adding Your Libraries

I added my existing anime library already in use with Plex as the first Jellyfin media source. Jellyfin scrapes metadata from online databases and requires properly structured filenames to match shows correctly.

Step 4 — File & Folder Naming Cleanup

Audited and corrected folder and file names to align with Jellyfin's metadata matching conventions. This included:

- Ensuring show titles were spelled accurately and matched their official database entries
- Verifying episode filenames followed a consistent naming format (e.g., Show Name S01E01)
- Reorganizing any misnamed files that prevented proper metadata scraping

Step 5 — Custom UI via CSS Theming

Explored the Jellyfin GitHub community for custom CSS themes that modify the default Jellyfin interface. Accessed Jellyfin's Display Settings and pasted custom CSS code directly into the designated field. Cycled through several options and settled on a theme that improved the visual experience significantly over the default layout. This gave Jellyfin a more streaming service-like experience.

Step 6 — Plugin Installation

Installed two community plugins via the Jellyfin plugin catalog:

- Media Bar Enhanced - provides a richer, more feature-complete media player control bar
- Intro Skipper - adds a dedicated 'Skip Intro' button to TV show episodes, a feature that requires a paid Plex Pass subscription on the Plex platform

Step 7 — Hardware Transcoding (Intel ARC B570 / QSV)

Enabled hardware transcoding in Jellyfin's playback settings, selecting Intel Quick Sync Video (QSV) as the transcoding method. This offloads video transcoding from the CPU (Ryzen 5 5500) to the dedicated Intel ARC B570 GPU, resulting in better performance and lower CPU utilization during streaming.

Note: Hardware transcoding configuration is ongoing. Further tuning and testing is planned as experience with QSV settings grows.

Step 8 — Remote Access via Tailscale VPN

Configured remote access using Tailscale, an open-source, peer-to-peer VPN solution. Setup involved:

- Creating a Tailscale account
- Installing the Tailscale client on all relevant devices (home server + mobile devices)
- Logging each device into the shared Tailscale network
- Using the server's Tailscale-assigned IP address as the Jellyfin connection URL from external devices
- Successfully connecting to and streaming from the Jellyfin library over an iPhone 16 Pro on a cellular (non-Wi-Fi) connection

Step 9 — Ongoing Evaluation

Jellyfin is now running in parallel with Plex for active comparison. Evaluation focuses on streaming stability, transcoding performance, client app experience, and any feature gaps discovered during day-to-day use.

Plex vs. Jellyfin: Feature Comparison

The table below summarizes the key differences observed between Plex (free tier) and Jellyfin based on this project:

Feature	Plex Media Server	Jellyfin
Cost	Free (limited features) / Plex Pass \$6.99/mo \$249 for a lifetime pass	Completely free, open-source
Hardware Transcoding	Plex Pass required	Free (Intel QSV with Arc B570)
Skip Intro Button	Plex Pass required	Free via Intro Skipper plugin
Custom UI / CSS	UI is already very solid	Full custom CSS via GitHub themes
Remote Access	Plex relay (limited free)	Tailscale VPN (open-source)
Plugin Ecosystem	Limited	Active community plugins
Metadata Scraping	Automatic	Automatic (requires clean filenames)
Mobile App	Free	Free, no ads

Skills Demonstrated

- Windows server software deployment and configuration
- Media server setup and library management (metadata, file naming conventions)
- Hardware transcoding configuration — Intel ARC / Intel QSV (Quick Sync Video)
- Open-source plugin installation and ecosystem navigation
- Custom CSS UI theming via community GitHub resources
- VPN-based remote access configuration (Tailscale)
- Cross-platform testing and troubleshooting (Windows server, iOS client)
- Comparative technology evaluation and documentation

Key Takeaways

Jellyfin delivered on its promise of being a fully-featured, cost-free alternative to Plex. Features that would require a Plex Pass subscription hardware transcoding, skip intro, and advanced UI customization were available out of the box or via free community plugins.

The open-source nature of Jellyfin also means there is no reliance on a third-party company's infrastructure for remote access or account management. Tailscale further strengthens this by providing secure, direct-connection VPN access without exposing the server publicly.

The main tradeoff is that Jellyfin requires more manual configuration and community research compared to Plex's more polished onboarding experience. You have to go into this knowing that this isn't a plug-in and play situation, nor an 'it just works' ordeal. However, for a home lab environment focused on learning and cost efficiency, this is a worthwhile trade.

Next Steps

- Continue tuning Intel QSV hardware transcoding settings for optimal performance
- Test additional Jellyfin plugins and themes
- Monitor long-term stability and compare streaming reliability with Plex
- Document any additional configuration changes as the setup evolves